Week 11 Homework — Multi-Omics Paper Analysis

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Note on scope. The assignment permits any three of the four papers. All four are answered below, because they deliberately represent four different ways of making multi-omics data talk to each other: (i) proteome ↔ transcriptome ↔ spatial proteome with protein-level validation; (ii) metagenome ↔ metabolome ↔ clinical metadata with a predictive model; (iii) host transcriptome ↔ 16S microbiome across three diseases with sparse multivariate statistics; (iv) a genetically encoded metabolite sensor ↔ CRISPR screen ↔ epigenome. Answering all four also makes the cross-paper comparison in Section 5 possible. Each question is answered in the order "what was measured → which computational step and with which parameters → which figure carries the evidence".

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- **Paper 1** — Klingelhuber et al., *Nature Metabolism* 2024 · Q1 proteomic workflow and QC · Q2 PCA, temporal clustering, pathway enrichment · Q3 transcriptome ↔ spatial proteome integration and C19orf12
- **Paper 2** — Oh et al., *Cell Metabolism* 2020 · Q1 metagenomic and metabolomic workflow · Q2 random-forest modelling and multi-cohort ROC/AUC · Q3 pathway–metabolite–clinical integration
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- **Section 5** — cross-paper comparison of the four integration strategies, followed by **References**

Paper 1 — A spatiotemporal proteomic map of human adipogenesis

Klingelhuber F., Frendo-Cumbo S. et al. *Nature Metabolism* 6, 861–879 (2024). doi:10.1038/s42255-024-01025-8

The study asks how a preadipocyte becomes a lipid-storing adipocyte, and answers with three layers of the same system: an abundance time course (total proteome), a localization layer (protein correlation profiling, PCP), and a reference layer (primary adipocytes and adipose tissue plus public transcriptomes).

Q1. Analysis workflow: generation, quality control, quantification and analysis of the proteomic data

1. Design and sample generation. Four human adipogenesis models were differentiated and sampled at day 0, 2, 4, 6, 10, 12 and 14, n = 3 per model per time point, plus an early series (0 h, 2 h, 12 h, day 1) to catch the first response (Fig. 1a; Extended Data Fig. 1c,d). Three primary sample types from seven donors — preadipocytes (pAC), stromal-vascular fraction (SVF) and intact white adipose tissue (WAT) — were profiled in parallel as the in vivo reference. Peptides (500 ng) were run as 1-hour single shots in data-independent acquisition (DIA) mode; DIA was chosen explicitly for higher identification rates and fewer missing values than DDA.

2. Identification and quantification. Spectronaut (directDIA) searched against UniProt human UP000005640 with trypsin, peptide length 7–52 aa, two missed cleavages, carbamidomethylation fixed and methionine oxidation/N-terminal acetylation variable, and a 1% q-value cut-off at both precursor and protein level. Label-free intensities were log2-transformed, proteins were kept if they had at least two valid values in at least one condition, and remaining missing values were imputed from a normal distribution (downshift 1.8, width 0.3). Absolute copy numbers were obtained with the proteomic-ruler approach (histone-based), which also served as an independent normalization to test whether results depended on LFQ scaling.

3. Quality control, with the numbers that were actually reported. 8,268 protein groups in the complete dataset (5,979–7,061 per cell model; 3,638–6,403 in the primary samples); 86% of proteins identified from unique peptides; five orders of magnitude of signal dynamic range; 98% of peptides with 3–8 data points per peak; median protein CV 12.3%, with 72% of CVs below 20% and 42% below 10%; mean Pearson correlation between replicates 0.97. One sample failed QC and was excluded from the Extended Data Fig. 1 analysis.

4. Statistics on abundance. Significant change was called by one-way ANOVA with permutation-based FDR < 0.01 per model (two-condition comparisons by two-sided t-test, same FDR), giving 38% of the proteome remodelled in all four models; 6–14% of proteins changed more than tenfold or were exclusive to one state. Set logic then produced the nested core: 6,569 proteins significant in at least one model, 2,403 in all four, 3,939 in at least three of four, and 1,979 with a conserved temporal trajectory.

5. Localization layer (PCP). Cells were mechanically lysed and separated on an 11 mL continuous 20–55% sucrose gradient (100,000 g, 3 h, SW40 Ti), 0.5 mL fractions were collected, and each fraction was measured by the same DIA method (3,500–5,600 proteins per fraction). Each protein's profile was scaled 0–1 across fractions, averaged over biological replicates, rescaled, and profiles that correlated negatively between any replicate pair were discarded.

6. Organelle assignment. An SVM with RBF kernel (Sigma 0.2, C 8), implemented in Perseus, was trained on marker proteins from three independent sources (a HeLa organelle map, the Protein Atlas, and LDKP for lipid droplets). Four adipocyte replicates, three preadipocyte replicates and one hAPC replicate were concatenated for classification. Result: 5,675 (preadipocyte) and 5,530 (adipocyte) proteins localized, with mean prediction accuracy 96% and 88% respectively (Fig. 2b–e; Extended Data Fig. 5c). Secondary

compartments were inferred from the correlation of a profile with in silico mixtures of pure organelle profiles; peroxisomes were excluded because they co-float with the ER in preadipocytes.

7. Enrichment and clustering. One-sided Fisher's exact tests on UniProt keywords, GO, KEGG and CORUM, with enrichment score > 2 and Benjamini–Hochberg FDR < 0.1; clustering by supervised Euclidean hierarchical clustering of z-scored temporal profiles on shared time points.

8. Validation of localization by an orthogonal method. Selected assignments were checked by immunofluorescence (SLC25A10, BCAT2, ALDH1L1/ALDH1L2 with TOM20 co-staining, C19orf12), which is what makes the PCP predictions usable rather than merely plausible.

Q2. Data interpretation: PCA, temporal clustering and pathway enrichment used to identify conserved adipogenic programmes

1. PCA as a trajectory test, not a decoration. PCA was run on proteins with at least two valid values across all models and primary samples. PC1 (24.7% of variance) and PC2 (13.8%) order samples along differentiation; the cell models project from the preadipocyte position toward the mature adipocyte position, and the primary pAC/SVF/WAT samples fall onto the same axis (Fig. 1c). That is the first piece of evidence that the in vitro time course reproduces in vivo adipogenesis rather than a culture artefact.

2. Interpreting the axes instead of only plotting them. PC1 loadings are dominated by lipid-droplet and lipogenic proteins (PLIN1, PLIN4, ADIPOQ, FABP4, SCD, ACSL1, MGLL, LIPE, ABHD5, ACACB), so PC1 can be read as "lipogenic maturation". When each model is analysed alone, PC1 remains the differentiation axis (21–36% of variance), and PC3/PC4 (6.9%/5.2%) capture model identity, with hWA sitting at a less mature position — a statement that is corroborated independently by microscopy and marker mRNA/protein levels.

3. Defining "conserved" operationally. Conservation was not defined by magnitude but by shape: after z-scoring and dropping time points not shared by all models, a protein was called conserved only if its temporal profile correlated positively in every pairwise comparison between the four models. Of 3,939 proteins significant in at least three models, 1,979 passed. This is the key methodological decision of the paper, because it separates "changed in this cell line" from "changed in human adipogenesis".

4. Clustering gives the temporal programme. Supervised hierarchical clustering of those 1,979 profiles produced three dominant phases. Early: cell cycle and protein biosynthesis down, lysosome and glycosaminoglycan degradation up. Intermediate: fatty-acid metabolism, TCA cycle, respiratory chain and ATP synthesis up; DNA replication, cytoskeleton, cell adhesion and WNT down. Late: cholesterol biosynthesis and lipid-droplet proteins up, spliceosome and mRNA processing down (Fig. 1g,h).

5. Enrichment converts clusters into pathways. Each cluster was annotated by Fisher's exact test against GO/KEGG/keywords (enrichment score > 2, BH FDR < 0.1), which is how the phase names above are justified rather than asserted (Fig. 1h). Pathway-level analysis was then repeated on the mitochondrial subset and cross-referenced to localization, which is where the branched-chain amino acid and one-carbon stories come from.

6. Two controls that make the clustering credible. (i) The 1,500 proteins unchanged in all four models are enriched for transcription factors, chromatin modifiers and plasma-membrane transporters — i.e. the temporal clusters are not a generic "abundance noise" artefact. (ii) The whole pipeline was repeated on proteomic-ruler copy numbers, and the same clusters and enrichments were recovered, so the programmes do not depend on LFQ normalization.

7. What the conserved programme means biologically. Three phases of coordinated remodelling, in which the same cells first shut down growth, then build the machinery of lipid synthesis and oxidation, then accumulate lipid-droplet and cholesterol-biosynthesis capacity. 38% of the quantified proteome participates,

and half of the conserved core follows an identical trajectory in four genetically distinct models — which is the paper's actual claim about universality.

Q3. Multi-omics integration: transcriptome ↔ spatial proteome, and the evidence chain for C19orf12

1. Proteome–transcriptome pairing done at the same-model level. The hAPC time course was matched to a published transcriptome time course of the same cell model, deliberately avoiding cross-model comparisons. Correlation was computed with Spearman's rho (transcript normalized log2 counts vs protein log2 LFQ): 0.39–0.48 at individual time points, increasing at later stages (Extended Data Fig. 4a). The modest absolute value is the finding, not a nuisance: it quantifies how much of the adipogenic programme is post-transcriptionally controlled.

2. Quantifying discordance rather than hiding it. Protein and mRNA z-profiles were clustered side by side, showing that protein changes lag mRNA changes and that the delay is protein-specific (Extended Data Fig. 4b,c). 16% of temporal protein profiles correlated negatively with their mRNA. One-dimensional annotation enrichment of the correlation values then showed a pathway-level structure: metabolic pathways correlate well, whereas signalling, chromatin regulation, transcription regulation and protein-complex components correlate poorly (Extended Data Fig. 4f) — i.e. exactly the classes where post-transcriptional control dominates.

3. A formal rule for combining abundance and localization. Organelle proportion at each time point was computed as (first SVM organelle assignment from PCP) × (summed LFQ intensity from the total-proteome time course), using only proteins with the same assignment at intermediate time points (Fig. 3b; Extended Data Fig. 6d). The result is a quantitative statement about organelle composition over time — mitochondrial, ER, endosomal and lipid-droplet fractions rise; cytosolic and nuclear fractions fall — converging on the composition of primary adipocytes. Localization itself is dynamic: 1,323 proteins change their first organelle assignment, 654 of them with high confidence in both states, i.e. about 20% of the mapped proteome relocalizes during differentiation.

4. Two mechanistically interpretable relocalizations. Branched-chain amino acid catabolism is reorganized both in abundance and in space: BCAT1 is cytosolic and falls, while BCAT2 rises and moves from cytosol into mitochondria, matching the mitochondrial BCAA enzymes that all increase (Fig. 4b,c; Extended Data Fig. 7). One-carbon metabolism splits the other way: the mitochondrial branch decreases and the cytosolic branch increases, and ALDH1L1/ALDH1L2 relocate from mitochondria to cytosolic protein complexes (Fig. 4d–h; Extended Data Fig. 8). Localization changes were confirmed by immunofluorescence with TOM20 co-staining.

5. C19orf12: from candidate to function, in six steps.

(i) *Prioritization by three filters*: 59 lipid-droplet proteins were shared by both white-adipocyte models, 34 were adipocyte-exclusive relative to LDKP non-adipocyte datasets, and 29 of the 34 were significantly regulated in the core time course; C19orf12 additionally had the largest mRNA/protein induction and a conserved trajectory in all four models (Fig. 6a; Extended Data Fig. 10a–c).

(ii) *Localization*: its PCP profile places it in lipid-droplet/cytosolic fractions with a minor Golgi pool; immunofluorescence shows ring-like staining around lipid droplets; co-staining with TOM20 shows colocalization with mitochondria adjacent to droplets (Fig. 6b,c,e,f).

(iii) *Interactome*: C19orf12-GFP co-immunoprecipitation with DIA proteomics (n = 4, FDR < 0.05) pulled down TOM40 — the channel-forming subunit of the TOM complex — TOM22 and TOM5, plus C12orf75 and GOLGA4. The interpretation is a protein that simultaneously binds lipid droplets and the mitochondrial import machinery at LD–mitochondria contact sites.

(iv) *Loss-of-function phenotype*: early knockdown increases lipid-droplet number per nucleus, halves basal lipolysis and increases adiponectin secretion by about 50%; late knockdown decreases insulin-stimulated lipogenesis and basal lipolysis, arguing for impaired lipid turnover rather than impaired differentiation (Fig. 6g–j).

(v) *Mechanistic discrimination by bioenergetics*: maximal oxygen consumption falls after knockdown with or without exogenous palmitate, but the difference disappears when CPT1 is inhibited with etomoxir, while isoprenaline-stimulated lipolysis increases. Together these exclude a general mitochondrial defect and place the lesion specifically in mitochondrial long-chain fatty-acid utilization — C19orf12 sets the capacity of droplet-associated mitochondria to oxidize fatty acids, and its loss diverts fatty acids into storage (Fig. 6k–n).

(vi) *Human relevance*: MAGMA gene-set analysis flagged the protein against adiposity phenotypes, and in seven public adipose-tissue cohorts (784 participants, 171 male / 613 female) C19orf12 expression correlated inversely with body fat, adipocyte cell volume, plasma triglycerides and HOMA-IR, with no sex-specific effect in a meta-analysis ($p = 0.91$ common effects, $p = 0.98$ random effects) (Fig. 6o).

6. Why this counts as integration and not as decoration. The transcriptome layer supplies the temporal context and the post-transcriptional caveat; the spatial layer supplies a physical address; the co-IP supplies the molecular partner; the Seahorse experiment supplies the function; the human cohorts supply the physiological direction. No single omics layer in the paper could have produced C19orf12 as a lipid-droplet/TOM-complex contact-site regulator of fatty-acid oxidation capacity.

Paper 2 — A universal gut-microbiome-derived signature predicts cirrhosis

Oh T.G., Kim S.M. et al. *Cell Metabolism* 32, 1–11 (2020). doi:10.1016/j.cmet.2020.06.005

The question is whether the gut microbiome carries a cirrhosis signal that survives geography, aetiology and host genetics. The design answers it by keeping the discovery cohort genetically and environmentally structured (probands plus their first-degree relatives) and then exporting the frozen model to cohorts from other continents.

Q1. Analysis workflow: shotgun metagenomics and metabolomics, from stool to disease-associated features

1. Cohort and phenotyping. 163 participants from the UCSD Familial Cirrhosis and Twins-and-Family cohorts: 27 NAFLD-cirrhosis probands and 54 non-NAFLD control probands, together with their first-degree relatives (38 and 44 respectively) (Table 1; Fig. 1A). Cirrhosis was established by biopsy (stage 4) or, in relatives, by imaging criteria — MRE ≥ 4.69 kPa for cirrhosis, ≥ 3.63 kPa for advanced fibrosis, or VCTE ≥ 11.8 kPa when MRE was contraindicated. Fat and fibrosis were quantified by MRI-PDFF and MRE, and the study reports an a priori power calculation (27 vs 54 detects AUC 0.88 at 80% power, alpha 0.01). The family structure is not incidental: it means controls share genetics, household and diet with cases, so a signal that survives is unlikely to be purely environmental.

2. Metagenomic library and read processing. DNA was extracted with the Qiagen MagAttract PowerSoil kit (EMP protocol), libraries prepared with quarter-scale NexteraXT, and sequenced on MiSeq with 1% PhiX. Reads were adapter-trimmed with skewer 0.2.2 in paired-end mode, human reads were removed by Bowtie2 against hg19, and the cleaned reads went into HUMAnN2 0.11.2 with MetaPhlAn2 2.7.7 for taxonomy. Function was assigned by mapping to the ChocoPhlAn pangenome and translating the unmapped fraction against UniRef90 with DIAMOND, then summarized onto MetaCyc pathways; abundances were renormalized with humann2_renorm_table and loaded through curatedMetagenomicData. Output: 1,216,335 gene families and 356 species, of which 310 annotated species (53 families, 115 genera) were retained across the 163 samples, plus 7,355 functional pathways and 188 strains.

3. Differential abundance. Taxonomic and functional features were tested with DESeq2; 39 of 7,355 pathways and 53 of 188 strains differed between cirrhosis and controls. The differentially abundant set was then used as the substrate for feature selection rather than being interpreted directly — an important choice, because it keeps the fold-change story and the predictive story separable.

4. Metabolomics. Stool was extracted in 50% methanol overnight (the solvent was benchmarked against 80% EtOH, water and EtOAc:MeOH:H2O by base-peak-chromatogram complexity), concentrated, and redissolved with internal standards. Separation used a Kinetex C18 column on a Dionex UPLC coupled to a Bruker MaXis qTOF (full scan m/z 50–2,000, top-10 CID, dynamic exclusion). Raw data were converted to .abf and processed in MS-DIAL 3.66 with explicit settings (MS1 tolerance 0.01 Da, MS2 0.025 Da, minimum peak height 1,000, smoothing level 3, minimum peak width 5 scans, alignment RT tolerance 0.2 min and MS1 0.015 Da), followed by manual removal of false-positive features. Annotation used MS-FINDER 3.20 with a stated confidence rule (mass error ≤ 10 ppm and MS2 matching score ≥ 8/10, relaxed to about 7.5–8.0 for the metabolites of interest). This yielded 435 quantified stool metabolites.

5. Diversity and ordination. Within-sample diversity was Inverse Simpson (phyloseq estimate_richness, t-test); between-sample structure was PCoA on weighted UniFrac with PERMANOVA (vegan adonis). Cirrhosis samples had lower alpha diversity ($p < 0.05$ in the UCSD cohort and reproduced in the Chinese and Italian cohorts) and separated from controls (PERMANOVA $p < 0.001$). Notably, alpha diversity in cirrhosis

correlated with LDL, prothrombin time, INR and insulin, which links the ecological change to liver function rather than to aetiology.

6. What the workflow deliberately does not do. No batch correction is applied between the discovery and validation cohorts; instead the model is retrained/transferred and the transformation (centring and scaling via caret preProcess) is applied to the external relative-abundance tables. This is a modelling decision that is discussed below, because it is also the weakest point of the validation.

Q2. Predictive modelling: how the signatures were selected, modelled and validated with random forests and ROC/AUC

1. Feature selection, two criteria in series. For each data type, features were first filtered by differential abundance (DESeq2) and then ranked by random-forest importance, using mean decrease in Gini as the ranking statistic. This combination is the reason the signatures are compact: 19 species out of 310, 17 metabolites out of 435, and 39 pathways.

2. Model fitting. Random forest was implemented with the train function of R's caret package, with 10-fold cross-validation via trainControl, on the 27 cirrhosis / 54 control proband set. No hold-out split within the discovery cohort is reported; the cross-validated AUC is therefore an internal estimate, and the external cohorts carry the real validation burden.

3. Performance of the single-omics models. The 19-species model reached AUC 0.91 (95% CI 0.89–0.93; Fig. 1F) and the 17-metabolite model reached the same AUC 0.91 (Fig. 3B). The two data types independently carry essentially the same discriminative information — an observation that is itself an argument that both are reading the same underlying dysbiosis.

4. Composition of the signatures. Up in cirrhosis: *Veillonella parvula* (the single most discriminatory adverse species despite low abundance), *Veillonella atypica*, *Ruminococcus gnavus*, *Clostridium bolteae*, *Acidaminococcus* sp. D21 (15-fold higher). Down: *Eubacterium eligens*, *Eubacterium rectale*, *Faecalibacterium prausnitzii* (the most important beneficial species). At strain level the extremes were *Acidaminococcus* sp. D21 (+15-fold) and *Dorea longicatena* (−1.3-fold); *Veillonella dispar* was 8.6-fold higher (adjusted $p = 4.79 \times 10^{-43}$). On the metabolite side the leading discriminators were glutamyl-phenylalanine isomer, leucine/phenylalanine derivatives, C18-sphingosine, glycochenodeoxycholic acid (a FXR ligand), increased indole/tryptophan species with reduced kynurenic acid, and reduced vitamin D derivatives.

5. Adding cheap clinical variables. Because the Chinese validation cohort only provided albumin, the authors tested the 19 species + age model plus serum albumin (a 22-feature model): AUC rose to 0.91 in the UCSD relatives cohort and 0.95 in the Chinese cohort (Fig. 5C,D). In the mixed-fibrosis cohort, adding AST — a marker that rises specifically in cirrhosis — raised discrimination of cirrhosis from NAFL from 0.85 to 0.94 and from stage 1–3 fibrosis from 0.84 to 0.91 (Fig. 5E–H). The sequence is informative: the microbiome alone detects cirrhosis, and routine serum markers are what let the model separate cirrhosis from earlier fibrosis.

6. Validation design, stated precisely. Internal: the relatives cohort, 51 non-NAFLD, 21 NAFLD without advanced fibrosis, 3 NAFLD-fibrosis and 7 cirrhosis, AUC 0.88 (0.72–1.00) (Fig. 5A). External, geographically and aetiologically distinct: a Han Chinese quantitative metagenomic cohort (123 patients and 114 controls) AUC 0.86, comparable to that cohort's own published 15-gene marker (0.84); an Italian 16S cohort (46 cirrhotic patients as stated in the text, 35 in the Fig. 5B legend, versus 14 controls; reads re-processed in DADA2 with RDP assignTaxonomy because raw fastq were available) AUC 0.89 (Fig. 5B). Mixed fibrosis cohort (biopsy-proven: 36 stage 0, 41 stage 1–3, 9 stage 4) AUC 0.85 and 0.84. Cross-disease generality is demonstrated for alcoholic liver disease, hepatitis B, hepatitis C and NAFLD. The honest limitation, stated by the authors, is that the validation cohorts lacked most clinical metadata, so the multi-feature model could not be tested with the same covariates everywhere.

7. Why this is a fair claim of "universal". The same 19 taxa, frozen from a San Diego twin-and-family cohort, discriminate cirrhosis in Chinese and Italian populations with different aetiologies, different sequencing platforms (shotgun vs 16S) and different depths. AUC degradation from 0.91 to 0.86–0.89 is the realistic cost of transport, and reporting it rather than a single discovery AUC is the point.

Q3. Multi-omics integration: microbial pathways, metabolites and clinical parameters

1. Pathogen-free statistical stitching at the feature level. The 19 species were correlated against the 17 metabolites by Spearman (19 × 17 matrix, $p < 0.05$ marked; Fig. 4A). This is a deliberately simple integration: it makes no causal assumption and simply asks whether the two signatures are describing the same patients.

2. Extension to gene families, and a network. Microbial gene families from the discriminatory species were added to the correlation analysis, producing a network of 33 nodes and 338 edges (species, gene families, metabolites) rendered in Cytoscape (Fig. 4B). This is where the integration becomes mechanistic in a weak sense: it identifies candidate enzymatic links (for example within *Ruminococcus gnavus*) rather than only co-occurrence.

3. Functional-pathway layer as the bridge between taxa and metabolites. HUMAnN2/MetaCyc pathways provided the middle layer: 39 pathways differed, concentrated in aromatic amino acid, branched-chain amino acid, fatty acid and nucleotide biosynthesis, and these pathways mapped onto *Acidaminococcus* and *Veillonella* — the same genera that dominate the species signature. The pathway layer is also what was compared across cohorts by RV coefficient (FactoMineR coeffRV): the top discriminatory pathway patterns were concordant between the current study, the Chinese cohort and the authors' earlier San Diego cohort (Fig. 2B). This cross-cohort concordance at pathway level is stronger evidence of universality than AUC alone, because it does not depend on the classifier.

4. Clinical/metadata layer. Of 310 species, 108 correlated with clinical metadata; for example *Veillonella parvula* correlated inversely with albumin and platelet count, whereas *Eubacterium eligens* correlated positively — the expected direction if the signature is tracking liver dysfunction. Metabolites were tied to the same axis (alpha diversity vs kynurenic acid, vitamin D3, enterolactone; Fig. 3E,F), and multivariate separation of the 17 metabolites was shown by OPLS (ropls; Fig. 3C).

5. Experimental closure of one edge of the network. *Ruminococcus gnavus* VPI C7-9 was cultured anaerobically with twelve concentrations of chenodeoxycholic acid (0.001–5,000 μg/ml) to mimic the raised bile acids of cirrhosis, and the supernatant was analysed by non-targeted LC-MS: C18-sphingosine and 1H-indole-3-carboxaldehyde increased with CDCA dose (Fig. 4C). This converts a correlation in a heatmap into a testable, direction-bearing mechanism (bile acid → microbial sphingolipid/indole production), and it is the single experiment that most strengthens the integration.

6. What the integration shows and what it does not. It shows a coherent cirrhosis axis: bile-acid-rich, low-diversity, *Veillonella*/*Acidaminococcus*-enriched, tryptophan- and BCAA-derivative-rich, vitamin-D-depleted, reproduced across three continents and multiple aetiologies, and it shows that this axis is quantitatively comparable to serum markers. It does not show causality, and the authors say so: the study is cross-sectional, the cohorts are small, compensated and decompensated cirrhosis were not separated, and the species could be consequences of portal hypertension and changed bile flow rather than drivers.

Paper 3 — Shared and disease-specific host gene–microbiome associations across human diseases

Priya S. et al. *Nature Microbiology* 7, 780–795 (2022). doi:10.1038/s41564-022-01107-x

The question is not whether the microbiome differs between health and disease, but whether the *host–microbe association structure* itself is shared or disease-specific. That requires paired sampling, per-cohort modelling, and a cross-disease comparison made at the level of results rather than of pooled data.

Q1. Analysis workflow: paired host RNA-seq and 16S rRNA data across CRC, IBD and IBS

1. Paired design. Three independent case–control cohorts of colonic mucosal biopsies, each individual contributing one 16S profile and one host transcriptome from the same tissue: 208 pairs in total — CRC 88 pairs (44 tumours and 44 patient-matched normals from 44 patients), IBD 78 pairs (56 patients: 34 Crohn's disease and 22 ulcerative colitis; 22 non-IBD controls), IBS 42 pairs (29 patients and 13 controls) (Fig. 1a; Extended Data Fig. 1). Pairing within the same biopsy is the design feature that makes host–microbe association interpretable at all.

2. Microbiome processing, cohort by cohort. 16S was V4 (IBD, IBS) or V5–V6 (CRC) on MiSeq 2 × 250 bp, and the pipelines were *not* harmonised: USEARCH v7 merge/trim plus QIIME v1.7.0 OTU picking (CRC), USEARCH v7.0.1090 plus UPARSE (IBD), and SHI7 trim/QC/merge with BURST alignment (IBS). Taxa were filtered to remove archaea, chloroplast, kit/reagent contaminants and soil/water taxa (Salter protocol), summarized at species (when present), genus, family, order, class and phylum, and then **all rank matrices were concatenated into a single feature matrix** so that a taxon can be selected at whatever rank carries the signal. Taxa were kept if they reached ≥ 0.001 relative abundance in ≥ 10% of samples. Retained taxa: 235 (CRC), 121 (IBD), 238 (IBS).

3. Host processing. CRC: FastQC 0.11.5, Trimmomatic (3-bp sliding window, minimum phred 16), HISAT2 to hg38, featureCounts (Subread 1.4.6); 87.11% mean alignment, mean 22.5 M reads. IBS: MAP-RSeq v2.0.0 (TopHat 2.0.12 on hg19, Subread 1.4.4). IBD: published HMP2 pipeline. Genes were restricted to protein-coding (biomaRt 2.37.4), kept if expressed in ≥ 50% of samples, variance-stabilized with DESeq2 1.14.1, and genes below the 25% quantile of variance were dropped. Retained genes: 12,513 / 11,985 / 12,429. The different aligner/reference combinations are an acknowledged source of heterogeneity and the reason all modelling is within-cohort.

4. Normalization. Microbiome data were transformed with the centred log-ratio (CLR). Two CLR strategies were compared — CLR of the concatenated multi-rank matrix versus CLR per rank then concatenation — and the first was adopted because the second produced multimodal distributions; agreement between them was high (mean Pearson 0.92, Spearman 0.87; Extended Data Fig. 5). Host expression was standardized after VST. No rarefaction and no batch-correction algorithm were used; instead each cohort was modelled separately, and cross-disease comparison happened only at the results level.

5. First question asked of the data: is there any global concordance? Procrustes analysis (vegan 2.4-5; Aitchison distance on host expression, Bray–Curtis on microbiome, 9,999 permutations) found host–microbiome concordance in CRC ($P = 0.0001$) but not in IBD ($P = 0.10$) or IBS ($P = 0.42$), confirmed by Mantel tests (0.0026 / 0.260 / 0.953). This negative result is what justifies feature-level sparse integration: a global correlation does not exist, so the signal must be sought in small subsets of genes and taxa.

6. Why cases and controls are modelled separately. Every association model was run in patients and in controls independently, and only associations found in patients but not controls were retained. As a control

for sample size, cases were downsampled to the control n and no overlapping associations appeared — i.e. the disease-specific associations are not an artefact of having more cases.

Q2. Integration algorithms: sparse CCA, LASSO regression and pathway enrichment

1. Sparse CCA for group-level (pathway-level) structure. Implemented in R 3.3.3 with PMA 1.1. The objective maximizes u^TX^TYv subject to $u^TX^TXu \le 1$, $v^TY^TYv \le 1$ and L1 bounds $\|u\|_1 \le \lambda 1$, $\|v\|_1 \le \lambda 2$ — i.e. canonical correlation with sparsity, so a component is driven by a few taxa and a few hundred genes rather than by the whole matrix. Because PMA's permutation-based tuning is one-dimensional, the authors wrote a two-dimensional grid search with leave-one-out cross-validation and selected the penalty pair with the highest CV correlation. Tuned values: CRC λ1 = 0.15, λ2 = 0.20; IBD 0.177 / 0.333; IBS 0.4 / 0.1 — note that the IBS solution is far sparser on the microbiome side, consistent with its smaller sample size.

2. Component significance without a permutation test. First 10 components were computed per cohort (deflation, so components are uncorrelated). Significance was assessed by leave-one-out: for each left-out sample, the component scores were recomputed from the remaining samples and the held-out scores correlated (cor.test), over all n held-out samples, with Benjamini–Hochberg FDR < 0.1 within cohort. Significant components: 7 (CRC; mean 828 genes and 8 microbes), 4 (IBD; 2,095 genes, 6 microbes), 6 (IBS; 577 genes, 61 microbes) (Fig. 1a; Supplementary Tables 2–4). The contrast is the biological result: IBD associates with many genes and few microbes, IBS with few genes and many microbes.

3. Pathway enrichment of component gene sets. Fisher's exact test with the sparse-CCA input genes as background, gene sets from MSigDB canonical pathways (KEGG + PID), excluding sets smaller than 25 genes, larger than 300 genes, or with fewer than 5 overlapping genes, FDR < 0.1, and each pathway assigned to its most significant component. Disease-specific pathways were then defined by a two-part rule: significant only in cases at FDR < 0.1, or significant in both but differentially enriched between cases and controls at FDR < 0.2 while still FDR < 0.1 enriched. Differential enrichment used the comparative log-odds statistic (CompGO-style z on log δ with its standard error), which corrects for the very different cohort sizes.

4. LASSO for gene-level resolution. For each host gene, a lasso regression was fitted with that gene's expression as response and about 200 taxa plus covariates (gender; CD vs UC; IBS-C vs IBS-D) as predictors — the rationale being that OLS is unstable when $p \gg n$ and only a few taxa should matter per gene. Implementation: glmnet 2.0-13, cyclical coordinate descent, λ chosen by leave-one-out CV. Inference on individual coefficients used the desparsified (regularized-projection) lasso with hdi 0.1-7, yielding bias-corrected coefficients, 95% CIs and p-values, with BH FDR < 0.1.

5. Stability selection as the specificity filter. Because a single lasso fit is unstable, stability selection (stabs 0.6-3) was run: random subsample of size n/2, refit with a perturbed λ near the CV optimum, repeated K = 100 times, selection frequency f per microbe computed and taxa retained at f ≥ 0.6 (following Meinshausen and Bühlmann, which also controls family-wise error). The reported association set is the **intersection** of stability-selected taxa and lasso-significant taxa (FDR < 0.1), with host gene–gender and host gene–subtype edges removed. Final counts: 755 (CRC), 1,295 (IBD) and 441 (IBS) host gene–taxa associations, derived from 745 genes × 120 microbes, 1,246 × 56 and 436 × 102 (Fig. 3a). Compute cost was about 12,300 gene-wise models per cohort, roughly 5.5 hours on five nodes.

6. Enrichment of the gene-level output and network rendering. Genes associated with microbes in patients were tested by Fisher's exact test against MSigDB KEGG + PID + REACTOME, excluding pathways with > 85 genes, < 10 genes, or < 5 overlapping genes — a filter that removed 297 too-large, 299 too-small and most of the remainder (mean 1,186) for insufficient overlap, leaving a mean of about 99 pathways tested per cohort; BH FDR < 0.1. This gave 18 disease-unique pathways (4 CRC, 9 IBD, 5 IBS). Networks were drawn in Cytoscape 3.5.1 with host genes as circles, microbes as triangles, edge width proportional to Spearman rho (not the lasso coefficient, which is biased), and gene–pathway edges in grey.

7. Cross-disease comparison was done on results, not on pooled data. Overlap was quantified with the overlap coefficient $|X \cap Y|/\min(|X|,|Y|)$. About 84% of input host genes were shared across cohorts but only about 37% of input taxa — a direct measure of how much the microbiome feature space differs between studies. Among disease-specific *associated* genes, 80% were also present as input elsewhere; among disease-specific *associated* taxa, only 12%. In other words the "disease-specificity" of the microbiome signal is largely a property of which taxa were measurable, which is an honest and important caveat on the paper's own headline.

Q3. Biological interpretation: shared and disease-specific regulatory networks

1. The shared core. Three pathways were significant in all three diseases: oxidative phosphorylation (mitochondrial energy metabolism) and the RAC1 pathway (immune response and mucosal repair) are named in the text, and the third entry of the shared block in Fig. 2a is the ERBB1 downstream pathway; pairwise overlaps added the proteasome, focal adhesion, cell-adhesion molecules, T-cell receptor, integrin, HDAC class I, MET and MAPK-TRK pathways (Fig. 2a). Against this, 102 pathways were disease-specific (52 CRC, 25 IBD, 25 IBS) — for example integrin α6β1/β4 and integrin β1 in IBD, and B-cell receptor signalling and ribosome in IBS.

2. The shared core uses disease-specific microbes. Host genes common to components in ≥ 2 diseases were enriched for gut-epithelial immune pathways (VEGF, complement and coagulation cascades, cytokine–cytokine receptor interaction; Fisher, BH FDR < 0.1), yet the microbes attached to those genes were disease-specific. The worked example is the RAC1 pathway: the same gene set (CYBB, NCF1, MAP3K1, CYFIP2, IQGAP3, NCKAP1, PLCB2, ARPC1B/3, ARHGDIA, CFL1, BCAR1, BAIAP2, STAT3, CTNNB1, PAK2, CTNNA1, ACTB/ACTR2, CYBA) pairs with *Streptococcus*, Synergistales and GN02 in CRC, with *Granulicatella* and *Clostridium* sensu stricto 1 in IBD, and with *Bacteroides massiliensis*, *Bifidobacterium* and *Odoribacter* in IBS (Fig. 2b).

3. Microbe-centric and gene-centric networks. Three microbes — Peptostreptococcaceae, *Streptococcus* and *Staphylococcus* — associate with host expression in all three diseases, and 20 microbes do so in ≥ 2 diseases (Fig. 4a). Five host genes are shared by all three diseases, and 135 genes associate in ≥ 2 diseases (Fig. 4b). The networks illustrate the two kinds of dissociation the paper is arguing for: the same microbe with different host targets (Peptostreptococcaceae positively with PYGB and NCK2 in IBD — WNT/β-catenin and integrin/EGFR — but negatively with HAS2 in IBS and with GAB1 in CRC), and the same host gene with different microbes (PINK1 pairs with *Collinsella* in CRC, Peptostreptococcaceae in IBD and *Blautia* in IBS).

4. Disease-specific biology that is internally coherent. CRC: IL-10 signalling with *Anaerolineae* and TM7 negatively correlated with CXCL8/IL1RN and Bacteroidales positively with CCR2/FPR1, plus NOTCH1-in-cancer and Syndecan-1 with *Parvimonas*/*B. fragilis* — an inflammatory–tumorigenic axis consistent with CRC immunobiology. IBD: Peptostreptococcaceae with MAPK3 (thrombin/PAR signalling) and VIPR1, and integrin β1 with Peptostreptococcaceae, *Intestinibacter* and *Phascolarctobacterium* — a leucocyte-recruitment axis. IBS: *B. massiliensis* with PLA2G4A (arachidonic acid), PPAR signalling (PLA2G4A, ACSL3/4, FABP3, LPL) and ribosome/B-cell receptor pathways — a barrier/lipid-mediator axis. Each fits the known pathophysiology of the disease without being derived from it.

5. The conclusion the data support, and the one they do not. The supported conclusion is structural: shared host pathways recruit different microbial partners in different diseases, and shared microbes attach to different host genes — i.e. host–microbe association networks are disease-context dependent even when their components overlap. The unsupported conclusion, which the authors avoid, is directionality: the study is entirely correlational, there is no replication cohort, and a disease-transformed mucosa may itself select the same taxa. The 12% figure for shared associated taxa is the quantitative expression of this limitation.

Paper 4 — A transcriptional biosensor reveals mechanisms of α-ketoglutarate signalling to chromatin

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α-ketoglutarate (αKG) is a TCA-cycle intermediate and a co-substrate for the αKG-dependent dioxygenases that demethylate DNA and histones. Whether its concentration inside the nucleus is regulated, and by what, was not measurable — so the paper first builds the measurement, then uses it as a screen readout.

Q1. Experimental workflow: how the αKG biosensor was developed and integrated with a CRISPR screen

1. The sensor comes from cyanobacteria. NtcA is an αKG-responsive transcription factor that activates *glnA*/*glnN* under nitrogen starvation in *Synechocystis*; αKG binding to NtcA homodimers increases DNA affinity. Recombinant NtcA bound αKG with Kd = 2.0 mM by microscale thermophoresis (16-point titration to 50 mM, oxaloacetate as counterscreen, PALMIST/GUSSI analysis) — deliberately in the range of intracellular αKG (1.35 mM measured in U251 cells), so the sensor is not saturated in its working regime (fig. S1A–B).

2. Turning a bacterial repressor–activator into a mammalian reporter. NtcA has a bipartite architecture (DNA-binding domain plus αKG effector-binding domain, structure PDB 8H40) but no mammalian transactivation domain or nuclear localization. The response element was designed from a PRODORIC position-weight matrix for NtcA sites in the *glnA* (PCC 6803) and *glnN* (PCC 6803, PCC 6903) promoters, synthesized as single sites or five-repeat tiles with native or TRE-derived spacers, and placed upstream of a minimal CMV promoter driving GFP. The chimera was optimized in two rounds: a pairwise combinatorial screen of transactivation domains (VP64, Zta, VP16 monomers, others) and NLS configurations in HEK293 cells, ± 1 mM dimethyl-αKG for 48 h, with GFP-positive fraction by flow cytometry gated against an empty-vector baseline; then a localization/architecture round using cell fractionation and immunoblot to remove non-nuclear and truncated products, NetNES1.1 to remove cryptic nuclear export signals from VP64, and replacement of the c-myc NLS by two SV40 large-T NLSs. The selected architecture is EF1α-driven **VP64-NtcA-2×NLS_SV40** (Fig. 1C–E; fig. S1–S2).

3. Calibration and specificity controls. Dose–response to dm-αKG (0.3, 1, 3 mM, 48 h) gave a graded GFP shift; glutamine titration (0.1, 0.5, 2 mM, 72 h) showed that the sensor tracks intracellular αKG and that dm-αKG rescues glutamine-deprivation-induced loss of output (Fig. 2A–I). Two mutant sensors are carried through the whole study as specificity controls: **NtcA-R90E** (αKG-binding dead) and **NtcA-ΔDBD** (DNA-binding dead). Oncometabolite counterscreens ((R)-2HG, dimethyl succinate, IDH1-R132H ± vorasidenib) dissociate biosensor output from oncometabolite-driven hypermethylation. Compartment validation compared BCAT1 (cytosolic) with BCAT2 (mitochondrial) deletion, measured whole-cell αKG by LC-MS and nuclear αKG by the sensor, and rescued with sgRNA-insensitive cDNA — establishing that the sensor reports the nucleocytosolic pool (Fig. 2J–O). Cell-cycle synchronization showed that the G2/M-associated GFP increase occurs in both the functional and the inactive arm, i.e. it is a reporter-copy-number effect and is subtracted by the two-arm design.

4. Screen design. A custom "αKG Regulators" CRISPR-Cas9 deletion library targeting **127 genes** encoding proteins that directly synthesize, catabolize or transport αKG was transduced into U251 cells carrying either the functional (WT NtcA) or the inactive (R90E) biosensor, at MOI ~0.3, 3×10^6 cells per arm, three replicates per arm, puromycin selection, and FACS at day 10 (Fig. 3A; data S1).

5. The two-arm readout is the core methodological idea. For each replicate of each arm, the top 10% and the bottom 10% of GFP-expressing cells were sorted separately, sgRNA cassettes were amplified from

genomic DNA and sequenced, and gene-level scores were computed as the ratio of sgRNA counts in the high versus low GFP bins with MAGeCK and the Broad GPP Apron pipeline. A hit had to show selective enrichment/depletion in the functional arm but **not** in the inactive arm — which removes every sgRNA effect that acts on fluorescence, viability or proliferation rather than on αKG (Fig. 3B,C).

6. Four hits, both signs. BCAT1 and DLD deletion increased nuclear αKG; GPT2 and SLC25A11 deletion depleted it (Fig. 3B). Each hit was validated with two independent sgRNAs by immunoblot and by GFP mean fluorescence intensity in both arms, by sgRNA-insensitive cDNA rescue, and by comparison of whole-cell versus nuclear αKG — which is what distinguishes a genuine compartment effect from a change in total metabolite (Fig. 3D–K; fig. S6, S8). Non-hits were examined explicitly: BCAT2, GLUD1, IDH3A, OGDH, PDHA1 and SNRPG either had no effect or were confounded by toxicity.

7. What the screen buys that a metabolomics survey cannot. An LC-MS comparison of 127 knockouts would be 127 experiments; the sensor converts the same question into one pooled selection. It also localizes the readout: two of the four hits (GPT2, SLC25A11) are mitochondrial proteins whose loss changes *nuclear* αKG, which is the observation that generates the rest of the paper.

Q2. Multi-omics analysis: connecting αKG metabolism to chromatin with metabolomics, ChIP-seq and RNA-seq

1. Metabolomics with absolute quantification. Relative αKG was measured by LC-MS on a ZIC-pHILIC column with a Q Exactive HF-X or Exploris 480 (120,000/240,000 resolution, m/z 80–1,200, polarity switching), integrated in El-Maven 0.12.0, quantified in TraceFinder 5.2 SP1, and normalized to total ion counts. Absolute αKG used isotope dilution with a $^{13}C_4$-αKG internal standard at 200 nM, a 10-point calibration from 10 nM to 10 µM with $1/X^2$ weighting, and division by cell number × single-cell volume (Beckman VI-CELL-XR, spherical approximation) — giving 1.35 mM whole-cell αKG in U251. GC-MS (Agilent 7890B/5977A, MOX/TBDMS derivatization, Metran integration) served as an orthogonal steady-state check. Flux was measured with α-^{15}N-glutamine (4 mM, 2 h, ± 1 mM pyruvate) and AccuCor natural-abundance correction, reporting Ala(M+1)/Glu(M+1) as the GPT2 flux readout.

2. Epigenomic readouts, two resolutions. Global DNA methylation (5mC and 5hmC) was measured by enzymatic hydrolysis and UHPLC-MS/MS with explicit SRM transitions and expressed as 5mC:dC normalized to the ascorbate-treated control (n = 4). Locus-resolved methylation used the duet multiomics evoC chemistry at single-base resolution on 80 ng of brain DNA, with genome-wide comparison by Wilcoxon signed-rank test.

3. ChIP-seq with spike-in normalization. H3K4me3, H3K9me3 and H3K27me3 were profiled in P19 mouse brain with 24 ng of *Drosophila* spike-in chromatin and 2 µg of spike-in antibody per sample, 200–400 bp sheared chromatin, and 30/50/40 million reads per sample respectively (NextSeq 2000, PE100). Reads were processed with TrimGalore 0.6.10/FastQC 0.12.1, Bowtie2 2.5.1 to mm10, deduplicated with Picard 3.0, **normalized using the spike-in values and the input**, and peaks called with MACS2 2.2.9.1 (broad setting for H3K9me3 and H3K27me3, FDR < 0.1). Differential peaks between knockout and wild type were identified with DiffBind 3.16.0 in R 4.2, and distribution shifts were tested by Kolmogorov–Smirnov on metagene and tornado plots. The spike-in is what makes "global hypermethylation" a quantitative statement rather than a normalization artefact.

4. Transcriptomics and pathway statistics. RNA-seq of P19 brain was analysed with DESeq2-style volcano plots and **GSEA 4.3.3** with gene-set permutation and FDR ≤ 0.25, plus GO enrichment via geneontology.org.

5. The integration step is an intersection with a direction constraint. H3K27me3-hypermethylated peaks (DiffBind) were intersected with genes *down*-regulated in the knockout: 884 genes (Fig. 6D), enriched for nervous-system development, neurogenesis, gliogenesis, regulation of calcium concentration and locomotory behaviour — and GSEA of the Meissner brain H3K27me3 gene set gave NES = −2.07 (FDR = 0.001,

KO vs WT) falling to −2.17 when KO was compared with dm-αKG-rescued KO. In the opposite direction, H3K4me3-hypermethylated peaks intersected with *up*-regulated genes gave more than 1,200 genes. Requiring the epigenetic change and the expression change to have the sign predicted by the histone mark is the guard-rail that turns two genome-wide datasets into one mechanism.

6. Coherence test at the transcriptome level. The direction of change after dm-αKG treatment was compared between WT and KO transcriptomes by χ^2 (116 down/69 up in WT versus 343 down/494 up in KO; $p < 0.00001$), showing that dm-αKG selectively normalizes the knockout transcriptome. Locus-level tracks then connected the statistics to specific genes: *Hapln4*, *Etv4* and *Notch3* (H3K27me3 up, expression down) and *Eif4ebp1*, *Cux2*, *Nrg3* (H3K4me3 up, expression up) (Fig. 6J–O; fig. S15).

7. What the multi-omics analysis establishes. Not that αKG "regulates chromatin" in general — that is textbook — but the quantitative direction and the size of the effect: loss of the mitochondrial αKG route produces *global* H3K4me3/H3K9me3/H3K27me3 increases (all distribution shifts $p < 0.001$), a genome-wide but small shift in DNA methylation (single-base 5mC $p < 0.001$; bulk 5mC:dC not significant, $p = 0.23$), and a coherent transcriptional programme of neurodevelopmental gene repression.

Q3. Mechanistic validation: genetic perturbation, epigenomic profiling and αKG rescue establishing GPT2–SLC25A11

1. Ascorbate dependence defines the mechanism as dioxygenase-mediated. GPT2 or SLC25A11 deletion caused H3K4 and H3K9 hypermethylation only when 200 µM ascorbate-2-phosphate was present for 2.5 weeks — i.e. only when the αKG-dependent demethylases are catalytically active. Without ascorbate the mark does not change. This single experiment excludes "methylation changed because the cells were sick" and places the lesion in KDM activity.

2. Epistasis, in both directions. GPT2 overexpression raised nuclear αKG and demethylated histones **only when SLC25A11 was intact**, and failed to rescue SLC25A11-knockout hypermethylation (Fig. 4D; fig. S12A–E) — so SLC25A11 is downstream. Conversely, DLD deletion raised nuclear αKG in both GPT2-WT and GPT2-null cells and *rescued* the GPT2-loss methylation phenotype, with OGDH deletion phenocopying DLD (Fig. 4E–F; fig. S12H–M) — so GPT2 and DLD are non-overlapping nodes feeding the same pool. Two orthogonal perturbations with opposite signs, tested against each other, are what convert a pathway cartoon into an ordered mechanism.

3. Which GPT2 pool matters — the isoform experiment. GPT2 has a full-length MTS-containing isoform (ENST00000340124.9, dominant) and an MTS-less nucleocytosolic isoform (ENST00000440783.2), annotated by TargetP-2.0 and tested by isoform-specific overexpression plus immunofluorescence: only mitochondrial GPT2 supplies nuclear αKG (fig. S11A–G). This is the compartment argument that makes the SLC25A11 requirement necessary rather than coincidental.

4. Substrate supply closes the loop. Pyruvate (1 mM, 3 days) drives GPT2 flux toward alanine/αKG and raises nuclear αKG in WT but not GPT2-KO cells, and CSF pyruvate is about twice blood pyruvate — a physiological rationale for why a mitochondrial transaminase sets a nuclear metabolite pool in the brain.

5. In vivo: the same axis, with a rescue. Whole-body Gpt2 knockout mice (gene-trap, dying 17–30 days postnatally) showed pervasive H3K4, H3K9 and H3K27 hypermethylation in brain at P19, with H3K9 changes also in heart and H3K9/H3K27 in spleen and little change in liver, kidney, lung, stomach, colon or sciatic nerve (Fig. 5A; Fig. 6A–C). dm-αKG at 600 mg kg^{-1} i.p. daily from P0 reversed the histone hypermethylation (immunoblot, n = 5 brains per arm), normalized the transcriptome (χ^2 $p < 0.00001$), restored the H3K27me3/GSEA signature, and improved body weight (vehicle n = 10 vs dm-αKG n = 9, $p < 0.001$) (Fig. 5E–L). A pharmacological rescue in vivo by the same metabolite the genetic lesion removes is the strongest available form of causal evidence in a mouse model.

6. Generalization beyond the brain. In isogenic PIK3CA-mutant HCT116 xenografts, GPT2 loss caused the same H3K4/H3K9/H3K27 hypermethylation, rescued by dm-αKG, while vehicle-treated growth rates were equal — so the phenotype is not a developmental artefact of the knockout mouse, and it is not caused by a proliferation difference.

7. The mechanism, stated as a chain. Mitochondrial GPT2 converts glutamate plus pyruvate-derived amino-group acceptors into αKG; the oxoglutarate carrier SLC25A11 exports that αKG out of the mitochondrion into the nucleocytosolic pool; the nucleocytosolic αKG concentration sets the substrate availability for αKG-dependent KDMs (whose Km for αKG is 5–60 μM, well below the measured millimolar pool, so the system operates as a buffer with a large dynamic reserve); when the route fails, histone methylation rises globally and a neurodevelopmental gene programme is repressed. The paper's contribution is not the existence of this chain in the abstract but the demonstration that this particular pair of transporters is the rate-setting route, and that the nuclear αKG pool is regulated rather than merely equilibrated.

5. Cross-paper comparison

	Paper 1 (adipogenesis)	Paper 2 (cirrhosis)	Paper 3 (host–microbe)	Paper 4 (αKG–chromatin)
Primary data	DIA proteome time course; PCP spatial proteome; public transcriptomes	Shotgun metagenome; untargeted stool metabolome	Paired 16S + host RNA-seq, 3 cohorts	αKG biosensor + CRISPR screen; LC-MS/GC-MS; ChIP-seq; RNA-seq
Integration logic	Feature-level combination of abundance and localization, then orthogonal validation	Feature selection per omics type, model fusion, cross-cohort transfer	Sparse multivariate association within cohort, meta-comparison across cohorts	Sensor as a quantitative phenotype for a genetic screen; epigenome × transcriptome intersection
Statistical glue	ANOVA FDR 0.01; conserved-trajectory correlation; SVM (Sigma 0.2, C 8)	Random forest + 10-fold CV; ROC/AUC; RV coefficient for cross-cohort concordance	Sparse CCA (λ grid + LOOCV); desparsified lasso; stability selection K = 100	MAGeCK/Apron two-arm scoring; MACS2 + DiffBind on spike-in-normalized ChIP-seq; GSEA
Validation	Four models + primary tissue + 784-person cohorts; immunofluorescence	Three external cohorts (China, Italy) + mixed-fibrosis cohort	None (discovery only); cross-cohort overlap analysis instead	Mutant sensor arm, cDNA rescue, epistasis, mouse KO + dm-αKG rescue, xenograft
Strongest claim	1,979 proteins follow a conserved three-phase programme; C19orf12 sets mitochondrial fatty-acid oxidation capacity	19 taxa + age + albumin detect cirrhosis across continents and aetiologies	Same pathways and taxa recur, but their pairing is disease-specific	Mitochondrial GPT2 → SLC25A11 sets the nuclear αKG pool that constrains histone demethylation
Main limitation	Two-column in vitro systems; localization inferred statistically	Small cohorts; no causality	Correlational; no replication cohort; only 12% of associated taxa shared	Screen limited to 127 curated genes

What separates the four integration strategies. Paper 1 integrates by *address*: the same protein is measured both in abundance and in space, so the two layers multiply rather than being correlated. Paper 2 integrates by *prediction*: each omics layer is compressed to a signature, and the signatures are judged by whether they transfer to new cohorts — integration is validated externally rather than internally. Paper 3 integrates by *sparse association*: it never pools the cohorts, and the multi-omics claim is about the structure of the association matrix (which gene goes with which taxon), which is the only level at which a cross-disease comparison of heterogeneous datasets is defensible. Paper 4 integrates by *perturbation*: the metabolite measurement is made causal by turning it into a selectable phenotype, and the epigenome and transcriptome are then read as the downstream consequences of that perturbation.

A methodological point common to all four. In every paper the number that carries the argument is a control, not a highlight: the 1,500 invariant proteins in Paper 1, the 0.86–0.89 external AUCs in Paper 2, the cases-downsampled-to-controls comparison in Paper 3, and the inactive-sensor arm in Paper 4. Each converts a correlative multi-omics observation into something that could have failed, which is the difference between data integration and data display.

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